SPANDAN PATIL

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PROFESSIONAL SUMMARY

ML engineer and researcher with 3+ years of building production-grade AI systems at scale. Expert in LLMs, generative AI, and real-time ML infrastructure. Published researcher in graph neural networks and NLP. Built an MCP server at AWS Bedrock processing 70+ model variants and deployed conversational AI platforms serving 1,000+ daily users. Passionate about pushing AI capabilities from research to production.

WORK EXPERIENCE

Software Development Engineer Intern | Amazon Web Services (AWS) *New York, NY • May 2025 – Aug 2025*

- Engineered a production Model Context Protocol (MCP) Server on AWS Bedrock processing real-time operational metrics across 70+ ML model variants; reduced expert analysis time by 80% through natural language query interface
- Architected low-latency serverless pipeline (AWS Lambda + FastAPI) achieving sub-1-minute p90 response on complex analytical queries across distributed infrastructure
- Designed 20 specialized analytical MCP tools with trust-scoring algorithms and synthetic data pipelines for CloudWatch logs achieving 30% reduction in false-positive capacity alerts

Data Scientist II | Micron Technology *Hyderabad, India • Aug 2022 – Jul 2024*

- Architected RAG-based conversational AI platform integrating PaLM/Gemini LLMs with enterprise knowledge bases via LangChain; reduced report generation time by 3–4 hours per analyst across the organization
- Built high-frequency real-time monitoring system with 56 FastAPI endpoints and 43 BigQuery routines processing live manufacturing telemetry; cut mean time-to-detection by 30% weekly
- Deployed scalable ML inference stack (FastAPI + LangChain + React) with optimized model serving pipelines handling 1,000+ daily queries at <100ms p99 latency

RESEARCH & PUBLICATIONS

An Efficient Bet-GCN Approach for Link Prediction • ***IJIMIA 2023***

Graph Convolutional Networks for link prediction; achieved 95.08%/95.07%/95.3% accuracy on CORA, CiteSeer, and PubMed benchmarks

M*mds*: multimodal benchmark dataset for suspicious profile detection on twitter social network • ***Springer SNAM 2024***

Built 1M-tweet multimodal dataset; achieved 92.24% accuracy detecting suspicious social network profiles using NLP + ML

PROJECTS

Benchmarking LLM Reasoning Degradation Under Domain Shift Link

Benchmarked reasoning degradation across GPT-4o, Claude Sonnet, and Mistral on 200+ domain-shifted prompts across medical, legal, and financial contexts; quantified hallucination rate and confidence calibration gaps across models

News Sentiment to Intraday Price Movement Predictor Link

Fine-tuned FinBERT on 5,000 financial news headlines; built sentiment-to-price alignment pipeline using yfinance data across 50 S&P 500 tickers, demonstrating statistically significant next-day return predictability ($p < 0.05$) for high-confidence sentiment signals

SKILLS AND INTERESTS

ML/AI:	PyTorch, TensorFlow, LangChain, HuggingFace, OpenCV, Scikit-learn, Transformers/LLMs
Languages:	Python, Java, C++, JavaScript/TypeScript, SQL, Bash
Cloud/Infra:	AWS (Bedrock, Lambda, SageMaker, EC2), GCP (BigQuery), Docker, CI/CD (GitLab), FastAPI, React
Domains:	LLMs, Generative AI, Computer Vision, Graph Neural Networks, NLP, Real-time Systems, MCP

EDUCATION

M.S. Computer Science (AI/ML Focus) | New York University - Courant *New York, NY • Sep 2024 – May 2026*

- ***Key Coursework****: Deep Learning, Decision Models, Predictive Analytics, DevOps & Agile Methodologies, Building LLM Reasoners*

B.Tech. Electrical Engineering | NIT Jaipur, India *India • Aug 2018– Jun 2022*

- ***Final project:*** *Built RL-based dynamic pricing & scheduling algorithm for EV Charging Stations achieving 138.5% profit improvement*